Ecurious Challenge: How to Integrate a Smart Reservation System in a Charging App for Public EV Charging

— Ecurious

Build a charging app feature that lets EV drivers reserve a public charge point – and make sure the spot is actually free when they arrive. You do not need to be an e-mobility expert. The goal is to combine different perspectives—from software and UX to energy systems, hardware and business.

The Setup



Ecurious provides real reservation hardware installed at a charging bay, controllable via the Ecurious server. Your app connects to this server to reserve, release and monitor the bay. Public charging data (locations, connectors, power) comes from GoingElectric, one of the most comprehensive sources for public charge points in Europe.

- **Reservation hardware:** physically keeps the bay free for the driver who booked it
- **Built-in camera:** license plate recognition to identify which vehicle is standing in the bay
- **Ecurious server:** your access point to the hardware (reserve, release, status, plate events)
- **GoingElectric:** public charging information to build the map and search in your app

Your Mission

Create a working prototype that integrates a smart reservation system into a charging app for public EV charging – from finding a charger to arriving at a free bay.

The core integration (app ↔ Ecurious server ↔ reservation hardware) is mandatory. Then choose one or more areas to go deeper:

- **No-Shows:** Design how the system handles drivers who reserve but never arrive.
- **Blocked Bays:** Decide what happens when a car is already standing in the reserved bay.
- **License Plate Recognition:** Use the built-in camera to check that the right vehicle arrives, release the bay automatically and detect blocking vehicles.
- **Interoperability:** Make the reservation accessible to other charging apps and link it smartly.
- **New Use Cases:** Explore what else the reservation hardware enables – events, retail, hotels, fleets and more.
- **Utilisation:** Show how reservations can increase utilisation for Charge Point Operators (CPOs).
- **Business Case:** Develop a business model, including how to price reservations and no-shows.

What to Deliver

Your team should:

1. Build a charging app prototype (mobile or web) that shows public charge points based on GoingElectric data.
2. Connect to the Ecurious server and demonstrate a full reservation cycle on the provided hardware: reserve → arrive (vehicle check-in via license plate recognition) → charge → release.
3. Document the entire process: architecture, API calls, data flows, design decisions, tools used and what did not work.
4. Present your answers to at least two of the mission areas above.
5. Describe limitations and ideas for scaling the system to other apps, operators and locations.

A polished app is a bonus—not a requirement. We are looking for a thoughtful, well-documented and transferable approach.

Guiding Questions

Use these questions as inspiration – you do not need to answer all of them.

No-Shows

- How long should a reservation be held? Grace period, automatic release, reminders?
- How do you detect that the driver is not coming (GPS, ETA, check-in)?

- How do you price a no-show fairly without scaring drivers away?

Blocked Bays

- What if a car is already standing in the bay – a combustion car, an EV not charging, or an EV still charging?
- How does the system detect this (e.g. via the camera), and how is the waiting driver informed or re-routed?
- Who is responsible, and what is the escalation path for the operator?

License Plate Recognition

- How do you link a license plate to a reservation – and what if the driver arrives in a different car?
- How can the camera trigger actions automatically, e.g. releasing the bay or starting the charging session (plug & charge-like experience)?
- How do you handle recognition errors: dirt, darkness, foreign or temporary plates?
- How do you process plate data in a GDPR-compliant way (data minimisation, retention, consent)?

Interoperability

- How can other charging apps and e-mobility providers access the reservation (e.g. open API, OCPI, OCPP)?
- How do you link it smartly – deep links, roaming platforms, a shared booking standard?
- How is a reservation from app A recognised on site for a driver using app B?

New Use Cases

- How can the reservation hardware serve events, retail, hotels, workplaces or fleets?
- Can a retailer offer reserved charging as a customer benefit or loyalty perk?

Utilisation

- How can reservations increase utilisation instead of keeping bays empty?
- Which data and KPIs show the effect for Charge Point Operators?

Business Case

- Who pays: driver, CPO, retailer, app provider?
- How could a pricing model look – reservation fee, no-show fee, subscription, revenue share?
- What is the value of a guaranteed charging spot for the driver and for the operator?

Fair Play & Safety

Only use the hardware and server access provided for the challenge. Do not interfere with public or production charging infrastructure. Respect the terms of use of GoingElectric and any other data source you use. Treat user, location and license plate data responsibly and in line with GDPR.

Resources

- **GoingElectric:** public charging station directory and API – goingelectric.de/stromtankstellen
- **Ecurious server:** access credentials and API documentation (provided on site)
- **Reservation hardware:** installed test bay at the hackathon venue
- **Standards for inspiration:** OCPP (ReserveNow / CancelReservation) and OCPI (reservation commands)
- **Ecurious mentors:** available throughout the hackathon for questions

Assessment Criteria

The jury scores each team on five criteria from 1 (weak) to 5 (excellent). Scores are weighted to a total of 100 points. The core integration (full reservation cycle on the Ecurious hardware) is a prerequisite for the top ranks.

Criterion	Weight	What the jury looks for
Technical Implementation	30%	Working integration of app, Ecurious server and hardware; full live reservation cycle; robust handling of edge cases (no-show, blocked bay, plate recognition errors); code quality.
Solution Concept & Innovation	20%	Smart, original answers to the mission areas; creative use of license plate recognition; new use cases beyond public charging (events, retail, fleets).
Interoperability & Scalability	15%	How easily other charging apps and operators can connect (open API, OCPI/OCPP); transferability to other locations and hardware.
Business Case & CPO Value	20%	Plausible pricing for reservations and no-shows; clear value for drivers and Charge Point Operators; measurable effect on utilisation.
Documentation & Pitch	15%	Complete documentation of the process (architecture, API calls, decisions, what did not work); clear, convincing live demo and presentation.
Total	**100%**	

Tie-breaker: the higher score in Technical Implementation decides. Responsible handling of user and license plate data (GDPR) is expected from all teams; clear violations can lead to point deductions.

What You Can Win

The winning team will receive:

- **€1,000 in prize money**